

Tianyu Chen

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Education

Ph.D. Peking University, 2020 - 2025 (Supervisor: Prof. Tao Xie)

Research Focus: Software Testing, Software Analytics, and Intelligent Software Engineering
GPA: 3.82

B.S. Peking University, 2016 – 2020

Major in Computer Science and Technology
GPA: 3.77, Ranking: 17/247 (EECS, Peking University)

Shanghai High School, 2013 – 2016

Honorship

Nomination of Doctoral Dissertation Incentive Program in CCF TCSE (2025)

Outstanding Graduate of Peking University (2020)

5.4 Scholarship (2017, 2024)

Houston Alumni Association Scholarship (2018, 2019)

The 10th National College Student Mathematics Competition: First place

ICPC In Peking University: Second place (2023, 2024), Third place (2016, 2018, 2021, 2022)

Publications

* Equal contribution, [†] Corresponding author

Qinglin Zhu, **Tianyu Chen**[†], Shuai Lu, Lei Ji, Runcong Zhao, Murong Ma, Xiangxiang Dai, Yulan He, Lin Gui, Peng Cheng, Yeyun Gong, Pull Requests as a Training Signal for Repo-Level Code Editing, ICML'26

Ziyue Hua, **Tianyu Chen**[†], Yeyun Gong, Shuai Lu, Peng Cheng, Qinglin Zhu, Yibo He, Yingjie Fu, Wenpin Jiao, Wei Yang, Tao Xie, Synthesizing File-Level Data for Unit Test Generation with Chain-of-Thoughts via Self-Debugging, ASE'26

Shangdong Cao, Tianlei Song, Qiuping Yi, **Tianyu Chen**, Guoai Xu, Ningyu He, Haoyu Wang, CLIR: Liveness-Driven and Structure-Aware Fuzzing for the Cranelift Compiler, ISSTA'26

Tianyu Chen, Shuai Lu, Shan Lu, Yeyun Gong, Chenyuan Yang, Xuheng Li, Md Rakib Hossain Misu, Hao Yu, Nan Duan, Peng Cheng, Fan Yang, Shuvendu K. Lahiri, Tao Xie, Lidong Zhou, Automated Proof Generation for Rust Code via Self-Evolution, ICLR'25

Tianyu Chen, Zeyu Wang, Lin Li, Shang Ma, Ding Li, Xiaoning Chang, Pan Bian, Guangtai Liang, Qianxiang Wang, Tao Xie, Detecting Functionality-Specific Vulnerabilities via Retrieving Individual Functionality-Equivalent APIs in Open-Source Repositories, ECOOP'25

Tianyu Chen, Ding Li, Ying Zhang, Tao Xie, Jsmpo: Structural Deobfuscation of JavaScript Programs, TOSEM'25

Tianyu Chen, Lin Li, Bingjie Shan, Guangtai Liang, Ding Li, Qianxiang Wang, Tao Xie, Identifying Vulnerable Third-Party Libraries from Textual Descriptions of Vulnerabilities and Libraries, TOSEM'25

Tianyu Chen, Lin Li, Liuchuan Zhu, Zongyang Li, Xueqing Liu, Guangtai Liang, Qianxiang Wang, Tao Xie, VulLibGen: Generating Names of Vulnerability-Affected Packages via a Large Language Model, ACL'24

Dezhi Ran, Yingjie Fu, Yibo He, **Tianyu Chen**, Xing Tang, Tao Xie, Path Toward Elderly-Friendly Mobile Apps, IEEE Computer'23

Preprints

Tianyu Chen, Lin Li, Taotao Qian, Jingyi Liu, Wei Yang, Xueqing Liu, Guangtai Liang, Ding Li, Qianxiang Wang, Tao Xie, CompVPD: Iteratively Identifying Vulnerability Patches Based on Human Validation Results with a Precise Context

Technical Reports

Lei Qu, Lianhai Ren, Peng Cheng, Rui Gao, Ruizhe Wang, **Tianyu Chen**, Xiao Liu, Xingjian Zhang, Yeyun Gong, Yifan Xiong, Yucheng Ding, Yuting Jiang, Zhenghao Lin, Zhongxin Guo, Ziyue Yang, SIGMA: An AI-Empowered Training Stack on Early-Life Hardware, arXiv:2512.13488, 2025

Qingguo Hu, Zhenghao Lin, Ziyue Yang, Yucheng Ding, Xiao Liu, Yuting Jiang, Ruizhe Wang, **Tianyu Chen**, Zhongxin Guo, Yifan Xiong, Rui Gao, Lei Qu, Jinsong Su, Peng Cheng, Yeyun Gong, Sigma-MoE-Tiny Technical Report, arXiv:2512.16248, 2025

Other Research Experiences

2018 - 2020: (random algorithms & complexity)

Supervised by Prof. Hanpin Wang (Peking University) and Prof. Heng Guo (University of Edinburgh)

Completed a technical report titled "An improvement from degree 3 to degree 4," showing that the coloring number of a graph with degree 4 can be estimated by using the topical structure of the graph

Completed a technical report about the lower bound for the mixing time of random walk over matroids

Teaching Experience

Introduction to Computer Systems (2018.09 - 2019.01, 2019.09 - 2020.01)

Practice of Programming in C&C++ (2019.02 - 2019.06)

Data Structure and Algorithms (2020.09 - 2021.01)

Computer Networks (2021.02 - 2021.06)